

# SOFTWARE REQUIREMENTS SPECIFICATION

## BizMind BD

**AI-Powered SME Commerce, Procurement & Business Intelligence Platform**

*An Agentic AI-Enabled Web and Mobile Business Platform*

**Version 0.1 — Week 1 Requirements Baseline**

Prepared for  
Web Programming and Mobile Application Development  
Department of Computer Science and Engineering  
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## Document Control

| Field       | Value                                                 |
|-------------|-------------------------------------------------------|
| Project     | BizMind BD                                            |
| Document    | Software Requirements Specification                   |
| Version     | 0.1                                                   |
| Status      | Week 1 Requirements Baseline / Draft                  |
| Course      | Web Programming and Mobile Application Development    |
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## Revision History

| Version | Date        | Description                        | Status |
|---------|-------------|------------------------------------|--------|
| 0.1     | August 2026 | Initial professional SRS baseline. | Draft  |

## 1. Introduction

### 1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional, interface, data, AI, security and architectural requirements for BizMind BD. It establishes the requirements baseline for design, implementation, testing and future product versions.

### 1.2 Product Vision

BizMind BD is a Bangladesh-focused startup-oriented platform for small and medium-sized enterprises (SMEs). It combines core business management, B2B commerce, procurement, supplier intelligence and business analytics with an Agentic AI layer. The AI layer is intended to analyze business context, use approved tools and data, coordinate specialized agents and prepare recommendations or controlled actions for human approval.

### 1.3 Problem Statement

Many SMEs depend on paper records, spreadsheets, messaging applications and disconnected tools for sales, inventory, procurement and supplier communication. This creates poor visibility into stock, demand, supplier performance and purchasing needs. BizMind BD addresses this fragmentation through one platform that combines operational management with decision support and controlled automation.

### 1.4 Objectives

- Provide a unified web and Android platform for SME operations.
- Reduce manual effort in inventory and procurement.
- Provide business dashboards and actionable insights.
- Support demand forecasting and supplier evaluation.
- Demonstrate meaningful Agentic AI through coordinated, tool-using agents.
- Keep high-impact actions under human approval with auditability.
- Create a modular architecture that can evolve into a startup product.

## 1.5 Scope

Version 0.1 defines the overall product vision and requirements baseline. The initial implementable MVP will prioritize core SME management, procurement, analytics and at least one end-to-end Agentic AI workflow. Advanced ecosystem features are reserved for later versions.

## 1.6 Design Reference

The supplied Nagorik Sheba SRS v0.1 was used as a structural reference, particularly its professional cover, table of contents, detailed requirement identifiers, Priority column, architecture section, glossary and appendices.

filecite{turn2file0L11-L29}

## 2. Overall Description

### 2.1 Product Perspective

BizMind BD follows a client-server architecture with web and Android clients, an API/application layer, relational data storage, analytics/ML services and an Agentic AI orchestration layer. The separation of concerns allows new agents, integrations and product versions to be added without redesigning the entire platform.

### 2.2 Major Product Functions

- Authentication and role-based access
- Business and employee management
- Product/catalog management
- Inventory and stock movement
- Sales and customer management
- Supplier and supplier-offer management
- B2B procurement and purchase orders
- Expense tracking
- Business intelligence dashboards
- Demand forecasting
- Supplier intelligence
- Agentic AI business analysis
- AI-assisted procurement planning
- Human approval and audit trail
- Web and Android access

### 2.3 User Classes

| User Class             | Description        | Responsibilities                                                  |
|------------------------|--------------------|-------------------------------------------------------------------|
| Business Owner         | SME owner/manager  | Manage business, monitor KPIs, approve AI recommendations/actions |
| Employee               | Authorized worker  | Record sales, update inventory and perform assigned tasks         |
| Supplier               | B2B supplier       | Manage offers, prices and order information                       |
| Platform Administrator | System operator    | Manage platform-level accounts and monitoring                     |
| AI Agent System        | Internal component | Analyze data, invoke approved tools and prepare recommendations   |

## 2.4 Operating Environment

- Web: modern desktop/mobile browsers with responsive UI.
- Mobile: Android application for essential operations and monitoring.
- Backend: Node.js/Express or equivalent API services.
- Database: PostgreSQL target architecture.
- AI/ML: Python services for forecasting and analytics; LLM API/local model for agent reasoning.
- Communication: HTTPS/REST/JSON; push notifications where enabled.

## 2.5 Constraints

- Course schedule and team capacity require prioritization.
- External LLM/API availability and cost may affect AI functions.
- High-impact AI actions require human approval.
- Secrets and private business data must not be exposed to development AI tools.
- The architecture must remain modular for future versions.

## 2.6 Assumptions and Dependencies

- Forecasting requires sufficient historical transaction data; seeded/demo data may be used during development.
- Users have normal internet access for web/mobile operation.
- External AI/notification APIs may be used subject to availability.
- AI-generated content and code will be reviewed and tested by the team.

# 3. System Features and Functional Requirements

Requirement IDs follow FR-[MODULE]-[NUMBER]. Priority values are High, Medium or Low. High requirements are essential to the initial product direction; Medium requirements are important but can be deferred; Low requirements are enhancements or future-ready capabilities.

## 3.1 Authentication & Access Control

| ID         | Functional Requirement                                                                      | Priority | Purpose / Acceptance Intent |
|------------|---------------------------------------------------------------------------------------------|----------|-----------------------------|
| FR-AUTH-01 | The system shall allow users to register and log in using approved credentials.             | High     | Core access                 |
| FR-AUTH-02 | The system shall enforce role-based access for Owner, Employee, Supplier and Administrator. | High     | Authorization               |
| FR-AUTH-03 | The system shall support logout, session/token management and password recovery.            | High     | Account management          |
| FR-AUTH-04 | The system shall restrict business data to authorized users/businesses.                     | High     | Data isolation              |
| FR-AUTH-05 | The system shall record important authentication                                            | Medium   | Audit                       |

|  |                           |  |  |
|--|---------------------------|--|--|
|  | and authorization events. |  |  |
|--|---------------------------|--|--|

### 3.2 Business & Employee Management

| ID        | Functional Requirement                                                                                | Priority | Purpose / Acceptance Intent |
|-----------|-------------------------------------------------------------------------------------------------------|----------|-----------------------------|
| FR-BIZ-01 | The owner shall create and maintain the business profile.                                             | High     | Business setup              |
| FR-BIZ-02 | The owner shall add, deactivate and manage employees.                                                 | High     | Workforce                   |
| FR-BIZ-03 | The owner shall assign roles/permissions according to access policies.                                | High     | Access control              |
| FR-BIZ-04 | The system shall support business settings such as low-stock thresholds and notification preferences. | Medium   | Customization               |

### 3.3 Product & Catalog Management

| ID         | Functional Requirement                                                                   | Priority | Purpose / Acceptance Intent |
|------------|------------------------------------------------------------------------------------------|----------|-----------------------------|
| FR-PROD-01 | Authorized users shall create, edit, search, categorize and deactivate products.         | High     | Catalog                     |
| FR-PROD-02 | Products shall support name, category, SKU/code, unit, purchase price and selling price. | High     | Product data                |
| FR-PROD-03 | The system shall prevent deactivated products from new transactions.                     | Medium   | Consistency                 |
| FR-PROD-04 | The system shall support product search, filtering and sorting.                          | Medium   | Usability                   |

### 3.4 Inventory Management

| ID        | Functional Requirement                                                           | Priority | Purpose / Acceptance Intent |
|-----------|----------------------------------------------------------------------------------|----------|-----------------------------|
| FR-INV-01 | The system shall maintain current stock quantity for each product.               | High     | Core inventory              |
| FR-INV-02 | The system shall record stock-in, stock-out, adjustment and transaction history. | High     | Traceability                |
| FR-INV-03 | Approved sales and purchase receipts shall update stock consistently.            | High     | Integration                 |
| FR-INV-04 | The system shall generate                                                        | High     | Stockout prevention         |

|           |                                                                           |        |             |
|-----------|---------------------------------------------------------------------------|--------|-------------|
|           | configurable low-stock alerts.                                            |        |             |
| FR-INV-05 | Authorized users shall review current availability and inventory history. | Medium | Monitoring  |
| FR-INV-06 | The architecture should support future multi-location inventory.          | Low    | Scalability |

### 3.5 Sales & Customer Management

| ID         | Functional Requirement                                                                | Priority | Purpose / Acceptance Intent |
|------------|---------------------------------------------------------------------------------------|----------|-----------------------------|
| FR-SALE-01 | Authorized users shall record sales and sale items.                                   | High     | Sales                       |
| FR-SALE-02 | The system shall calculate line totals and sale totals automatically.                 | High     | Accuracy                    |
| FR-SALE-03 | Completed sales shall update relevant inventory quantities.                           | High     | Integration                 |
| FR-SALE-04 | The system shall maintain customer profiles when customer identification is provided. | Medium   | Customer data               |
| FR-SALE-05 | The system shall support sales search, filtering and date summaries.                  | Medium   | Reporting                   |

### 3.6 Supplier Management

| ID        | Functional Requirement                                                      | Priority | Purpose / Acceptance Intent |
|-----------|-----------------------------------------------------------------------------|----------|-----------------------------|
| FR-SUP-01 | Authorized users shall create and maintain supplier profiles.               | High     | Supplier data               |
| FR-SUP-02 | The system shall maintain supplier product offers, prices and availability. | High     | Procurement data            |
| FR-SUP-03 | The system shall maintain supplier performance indicators.                  | Medium   | Supplier intelligence       |
| FR-SUP-04 | The system shall support supplier search and comparison.                    | Medium   | Decision support            |

### 3.7 B2B Commerce & Procurement

| ID         | Functional Requirement                                           | Priority | Purpose / Acceptance Intent |
|------------|------------------------------------------------------------------|----------|-----------------------------|
| FR-PROC-01 | A business shall create purchase requests for required products. | High     | Procurement                 |
| FR-PROC-02 | Authorized users shall                                           | High     | Execution                   |



|            |                                                                               |      |                       |
|------------|-------------------------------------------------------------------------------|------|-----------------------|
|            | convert approved requests into purchase orders.                               |      |                       |
| FR-PROC-03 | The system shall compare supplier quotations/offers using available data.     | High | Cost-aware purchasing |
| FR-PROC-04 | The system shall track purchase-order status.                                 | High | Order tracking        |
| FR-PROC-05 | Received quantities shall update inventory after an approved receipt.         | High | Integration           |
| FR-PROC-06 | The architecture should support a future multi-business supplier marketplace. | Low  | Future version        |

### 3.8 Expense & Basic Financial Tracking

| ID        | Functional Requirement                                                         | Priority | Purpose / Acceptance Intent |
|-----------|--------------------------------------------------------------------------------|----------|-----------------------------|
| FR-FIN-01 | Authorized users shall record expenses with amount, category, date and note.   | Medium   | Expense records             |
| FR-FIN-02 | The system shall provide expense summaries by period and category.             | Medium   | Analysis                    |
| FR-FIN-03 | Expense indicators shall appear in business dashboards.                        | Medium   | BI                          |
| FR-FIN-04 | The prototype shall not claim to be a legally certified accounting/tax system. | High     | Scope control               |

### 3.9 Business Intelligence Dashboard

| ID       | Functional Requirement                                                     | Priority | Purpose / Acceptance Intent |
|----------|----------------------------------------------------------------------------|----------|-----------------------------|
| FR-BI-01 | The system shall display sales, inventory, procurement and expense KPIs.   | High     | Monitoring                  |
| FR-BI-02 | Owners shall filter dashboard data by relevant time periods.               | Medium   | Analysis                    |
| FR-BI-03 | The system shall identify selected trends and unusual changes.             | Medium   | Insights                    |
| FR-BI-04 | The dashboard shall use understandable charts, cards and tables.           | Medium   | Usability                   |
| FR-BI-05 | The architecture should support industry-specific KPIs in future versions. | Low      | Extensibility               |

### 3.10 Demand Forecasting

| ID       | Functional Requirement                                                                              | Priority | Purpose / Acceptance Intent |
|----------|-----------------------------------------------------------------------------------------------------|----------|-----------------------------|
| FR-AI-01 | The system shall estimate future product demand from historical sales where sufficient data exists. | High     | Forecasting                 |
| FR-AI-02 | The system shall support defined forecast horizons such as 7/14/30 days where feasible.             | Medium   | Planning                    |
| FR-AI-03 | The system shall warn when data is insufficient or confidence is low.                               | High     | Responsible AI              |
| FR-AI-04 | Forecast results shall be usable by the Procurement Agent.                                          | High     | Agent integration           |

### 3.11 Supplier Intelligence

| ID       | Functional Requirement                                                                                    | Priority | Purpose / Acceptance Intent |
|----------|-----------------------------------------------------------------------------------------------------------|----------|-----------------------------|
| FR-AI-01 | The system shall score suppliers using available factors such as price, delivery reliability and history. | High     | Supplier decision           |
| FR-AI-02 | The system shall compare eligible supplier offers for a procurement need.                                 | High     | Procurement                 |
| FR-AI-03 | The system shall show factors contributing to a supplier recommendation.                                  | Medium   | Explainability              |
| FR-AI-04 | The scoring model shall be replaceable without redesigning supplier master data.                          | Low      | Extensibility               |

### 3.12 Agentic AI Business Analyst

| ID          | Functional Requirement                                                                      | Priority | Purpose / Acceptance Intent |
|-------------|---------------------------------------------------------------------------------------------|----------|-----------------------------|
| FR-AGENT-01 | The Agent Orchestrator shall route business tasks to appropriate specialized agents.        | High     | Agentic core                |
| FR-AGENT-02 | The Business Analyst Agent shall analyze approved data for trends, risks and opportunities. | High     | Intelligence                |
| FR-AGENT-03 | Agents shall use approved tools/data for business-specific reasoning.                       | High     | Grounding                   |
| FR-AGENT-04 | Recommendations shall                                                                       | High     | Explainability              |

|             |                                                      |        |              |
|-------------|------------------------------------------------------|--------|--------------|
|             | include understandable supporting factors.           |        |              |
| FR-AGENT-05 | Important agent decisions/actions shall be recorded. | Medium | Auditability |

### 3.13 Agentic AI Procurement Workflow

| ID            | Functional Requirement                                                                  | Priority | Purpose / Acceptance Intent |
|---------------|-----------------------------------------------------------------------------------------|----------|-----------------------------|
| FR-PROC-AI-01 | The Procurement Agent shall receive a procurement need from a user or approved trigger. | High     | Task initiation             |
| FR-PROC-AI-02 | The agent shall retrieve inventory and sales/forecast information.                      | High     | Context                     |
| FR-PROC-AI-03 | The agent shall estimate a recommended purchase quantity.                               | High     | Planning                    |
| FR-PROC-AI-04 | The agent shall compare eligible supplier offers.                                       | High     | Selection                   |
| FR-PROC-AI-05 | The agent shall prepare product, quantity, supplier and rationale.                      | High     | Recommendation              |
| FR-PROC-AI-06 | High-impact purchase actions shall require authorized human approval.                   | High     | Human-in-loop               |

### 3.14 AI Assistant

| ID           | Functional Requirement                                                        | Priority | Purpose / Acceptance Intent |
|--------------|-------------------------------------------------------------------------------|----------|-----------------------------|
| FR-ASSIST-01 | Owners shall be able to ask supported business questions in natural language. | Medium   | Assistance                  |
| FR-ASSIST-02 | Data-specific answers shall retrieve authorized business data first.          | High     | Grounding                   |
| FR-ASSIST-03 | The assistant shall distinguish facts, estimates and recommendations.         | High     | Trust                       |
| FR-ASSIST-04 | The assistant shall refuse or clarify actions outside its permissions.        | High     | Safety                      |
| FR-ASSIST-05 | Bangla/English support should be considered in later iterations.              | Low      | Localization                |

### 3.15 Notifications

| ID | Functional Requirement | Priority | Purpose / Acceptance |
|----|------------------------|----------|----------------------|
|----|------------------------|----------|----------------------|

|             |                                                                                     |        | <b>Intent</b> |
|-------------|-------------------------------------------------------------------------------------|--------|---------------|
| FR-NOTIF-01 | The system shall generate low-stock notifications.                                  | High   | Operations    |
| FR-NOTIF-02 | The system shall notify relevant users about purchase-order status changes.         | Medium | Workflow      |
| FR-NOTIF-03 | The system shall notify owners about important AI recommendations requiring review. | Medium | Human review  |
| FR-NOTIF-04 | The architecture shall support future push/email/SMS channels.                      | Low    | Extensibility |

### 3.16 Android Mobile Application

| <b>ID</b> | <b>Functional Requirement</b>                                            | <b>Priority</b> | <b>Purpose / Acceptance Intent</b> |
|-----------|--------------------------------------------------------------------------|-----------------|------------------------------------|
| FR-MOB-01 | The Android app shall provide secure authentication.                     | High            | Mobile access                      |
| FR-MOB-02 | The app shall support essential product, inventory and sales operations. | High            | Mobile operations                  |
| FR-MOB-03 | The app shall display important notifications and AI recommendations.    | Medium          | Monitoring                         |
| FR-MOB-04 | The app should provide a concise owner KPI dashboard.                    | Medium          | Management                         |
| FR-MOB-05 | The mobile app shall use the same backend APIs as the web platform.      | High            | Consistency                        |

### 3.17 Audit, Approval & Governance

| <b>ID</b> | <b>Functional Requirement</b>                                                    | <b>Priority</b> | <b>Purpose / Acceptance Intent</b> |
|-----------|----------------------------------------------------------------------------------|-----------------|------------------------------------|
| FR-GOV-01 | The system shall record important user and AI actions with timestamp and source. | High            | Audit                              |
| FR-GOV-02 | The system shall distinguish AI recommendations from user-approved actions.      | High            | Transparency                       |
| FR-GOV-03 | High-impact AI-generated actions shall require explicit authorization.           | High            | Governance                         |
| FR-GOV-04 | The system shall preserve context needed to explain important recommendations.   | Medium          | Explainability                     |

|           |                                                                           |     |            |
|-----------|---------------------------------------------------------------------------|-----|------------|
| FR-GOV-05 | Administrators should be able to monitor selected system and AI activity. | Low | Monitoring |
|-----------|---------------------------------------------------------------------------|-----|------------|

## 4. External Interface Requirements

### 4.1 User Interfaces

- Owner dashboard: KPIs, alerts, procurement recommendations, approvals and business summaries.
- Employee interface: sales, inventory updates and assigned operational tasks.
- Supplier interface: profile, offers, quotations and order status.
- AI assistant: natural-language queries with recommendation rationale and approval controls.
- Android UI: simplified, touch-friendly workflows for essential operations.

### 4.2 Software Interfaces

| Interface            | Purpose                                              |
|----------------------|------------------------------------------------------|
| REST/JSON API        | Communication between web/mobile clients and backend |
| PostgreSQL           | Persistent relational business data                  |
| Python ML service    | Forecasting and analytics                            |
| LLM API / local LLM  | Reasoning/language for selected agents               |
| Git/GitHub           | Source control and collaboration                     |
| Notification service | Push alerts where enabled                            |

### 4.3 Communication Interfaces

- HTTPS/TLS for client-server and external APIs.
- REST/JSON for application communication.
- Authenticated service-to-service calls for AI/ML services.

## 5. Non-Functional Requirements

| Category        | Requirement                                                                                       | Priority |
|-----------------|---------------------------------------------------------------------------------------------------|----------|
| Security        | Protected APIs shall enforce authentication/authorization; credentials shall be securely handled. | High     |
| Privacy         | Business data shall be isolated by business and exposed only to authorized users.                 | High     |
| Performance     | Common CRUD operations should remain responsive under expected academic-demo load.                | High     |
| Reliability     | Core sales, inventory and procurement transactions shall preserve data consistency.               | High     |
| Scalability     | Architecture shall support more businesses, users, products, suppliers and agents.                | High     |
| Maintainability | Frontend, backend, data, AI and integrations shall be separated and                               | High     |

|                  |                                                                              |      |
|------------------|------------------------------------------------------------------------------|------|
|                  | documented.                                                                  |      |
| Usability        | Interfaces shall be understandable to SME users with basic digital literacy. | High |
| Explainability   | Important AI recommendations shall show supporting factors where feasible.   | High |
| Auditability     | Important user and AI actions shall be traceable.                            | High |
| Interoperability | Web and Android clients shall consume shared backend APIs.                   | High |

## 6. System Architecture Overview

BizMind BD follows a layered architecture. Users interact through Web and Android clients. Requests are handled by application services, which access business data and invoke analytics/ML or the Agent Orchestrator when intelligence is required. Specialized agents use approved tools/data and return recommendations. High-impact actions pass through human approval before execution.

|                                                                                                                 |                         |                               |
|-----------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------------|
| <b>USER / CHANNEL LAYER</b>                                                                                     |                         |                               |
| Business Owner • Employee • Supplier • Administrator                                                            |                         |                               |
| Web Application                                                                                                 | Android Application     | Future B2B / Partner Channels |
| <b>APPLICATION / API LAYER</b>                                                                                  |                         |                               |
| Authentication • RBAC • Business Services • Sales • Inventory • Procurement • Notifications                     |                         |                               |
| <b>INTELLIGENCE / AGENT LAYER</b>                                                                               |                         |                               |
| Agent Orchestrator → Business Analyst   Demand Forecasting   Supplier Intelligence   Procurement   AI Assistant |                         |                               |
| <b>DATA &amp; INTEGRATION LAYER</b>                                                                             |                         |                               |
| PostgreSQL / Business Data                                                                                      | Python ML / Forecasting | Approved External APIs / LLM  |
| <b>CONTROL &amp; GOVERNANCE</b>                                                                                 |                         |                               |
| Human Approval • Audit Log • Security • Monitoring                                                              |                         |                               |

Primary flow: User → Client → API/Application → Business Data/Analytics → Agent Orchestrator → Specialized Agent(s) → Recommendation → Human Approval → Approved Business Action → Audit/Notification.

### 6.1 Agentic AI Architecture

- Agent Orchestrator: selects and coordinates agents/tools.

- Business Analyst Agent: analyzes trends, anomalies, risks and opportunities.
- Demand Forecasting Agent: invokes forecasting models and interprets results.
- Supplier Intelligence Agent: evaluates suppliers and offers.
- Procurement Agent: combines inventory, demand and supplier data to prepare purchase recommendations.
- AI Assistant: converts supported natural-language questions into authorized data/analysis tasks.
- Tool layer: controlled access to inventory queries, sales summaries, forecasts, supplier scoring and procurement services.

## 6.2 Example End-to-End Workflow

Owner asks: “Do I need to purchase more cooking oil?” The system retrieves inventory and sales, invokes the Demand Forecasting Agent, sends the resulting need to the Procurement Agent, compares eligible suppliers through Supplier Intelligence, prepares a recommendation with rationale, requests owner approval and only then creates the purchase action.

## 7. Data Architecture

The target relational model is organized around clear ownership and transaction relationships. Detailed normalization and the ER diagram will be finalized in the design phase.

| Entity                            | Purpose                           |
|-----------------------------------|-----------------------------------|
| User                              | Identity/account information      |
| Business                          | SME organization/tenant           |
| Role / Employee                   | Business access and workforce     |
| Product / Category                | Catalog master data               |
| Inventory / Inventory Transaction | Stock and movement history        |
| Sale / Sale Item                  | Sales transactions                |
| Customer                          | Customer profile                  |
| Supplier / Supplier Offer         | Supplier data and offers          |
| Purchase Order / Item / Receipt   | Procurement lifecycle             |
| Expense                           | Expense records                   |
| Notification                      | Alerts                            |
| AI Recommendation                 | AI recommendation and rationale   |
| AI Action / Audit Log             | Traceability of important actions |

## 8. Security, Privacy & Responsible AI

- Role-based authorization shall be enforced at API/service level.
- Passwords, API keys and secrets shall never be committed to GitHub.
- AI agents shall access only approved tools and authorized business data.
- High-impact purchasing/financial actions shall remain behind explicit human approval.
- AI outputs shall be presented as recommendations/estimates where appropriate.
- Business data shall be isolated to prevent cross-business exposure.
- Synthetic or appropriately authorized data shall be used for development; private data should not be shared with public AI services without permission.

## 9. Version Roadmap & Upgrade Path

| Version | Focus | Planned Evolution |
|---------|-------|-------------------|
|---------|-------|-------------------|

|      |                           |                                                                                                |
|------|---------------------------|------------------------------------------------------------------------------------------------|
| 0.1  | Requirements Baseline     | SRS, scope, architecture direction and detailed requirements.                                  |
| 0.2  | Design Baseline           | Wireframes, ERD, API contracts, architecture/flow diagrams and GitHub Projects.                |
| 0.3  | Core Prototype            | Authentication, UI, CRUD modules and shared API.                                               |
| 0.4  | Intelligence Prototype    | Forecasting, supplier scoring, BI and first Agentic AI workflow.                               |
| 1.0  | Academic/Startup MVP      | Integrated SME operations, procurement intelligence, mobile support and end-to-end Agentic AI. |
| 1.1+ | Intelligence & Automation | Improved forecasting, anomaly detection and automated replenishment suggestions.               |
| 2.x  | B2B Network               | Supplier marketplace, quotations, order tracking and partner integrations.                     |
| 3.x  | Business Ecosystem        | Logistics, financing partnerships and industry-specific AI agents.                             |

## 10. MVP Acceptance Criteria

- Users can securely authenticate and access role-appropriate functions.
- A business can manage products, inventory, sales, suppliers and procurement.
- Transactions update inventory consistently.
- Owners can view meaningful KPIs.
- The system can produce a basic demand forecast when sufficient data exists.
- Supplier offers can be compared using defined factors.
- At least one end-to-end Agentic AI workflow can retrieve approved data, invoke intelligence, produce a recommendation and record the result.
- High-impact AI recommendations require human approval.
- Web and Android clients use the shared backend/API architecture.

## 11. Development & Traceability Plan

| Phase     | Deliverables                                                                         | Relevant SRS Areas           |
|-----------|--------------------------------------------------------------------------------------|------------------------------|
| Week 1    | Scope, objectives, roles, SRS, README, contribution guidelines, initial GitHub tasks | Sections 1–3                 |
| Week 2    | Wireframes, ERD, flowcharts, architecture refinement, GitHub Projects                | Sections 6–7                 |
| Weeks 3–4 | Authentication, navigation, basic UI, core structure, tests, Pull Requests           | AUTH/BIZ/PROD/INV            |
| Weeks 5–6 | CRUD, forms, APIs, testing and sprint reporting                                      | Core functional requirements |
| Later     | BI, forecasting, supplier intelligence, Agentic AI and approval workflow             | BI/AI/PROC-AI/GOV            |



## 12. GitHub & AI-Assisted Development

- Git/GitHub will provide version control, collaboration, Issues, Pull Requests, review and history.
- GitHub Issues/Projects will decompose requirements into assignable tasks and track progress.
- Feature branches will protect the stable main branch.
- Pull Requests will be used for peer review before important merges.
- AI tools may assist requirements, documentation, UI ideas, code scaffolding, tests and debugging; all output must be reviewed and validated by the team.
- No API keys, passwords, tokens, .env files or private business data shall be committed.

## 13. Acknowledgement

The development of BizMind BD is supported by software engineering, AI/ML and collaboration tools. These tools are used to improve development quality, documentation, teamwork and experimentation; they do not replace the team's responsibility for system design, implementation, testing and validation.

| Tool / Technology        | Purpose and Reason for Use                                                                                        |
|--------------------------|-------------------------------------------------------------------------------------------------------------------|
| Git & GitHub             | Version control, collaboration, Issues, Pull Requests, code review and development history.                       |
| GitHub Projects          | Task allocation, sprint planning, progress tracking and workflow visualization.                                   |
| AI / LLM Tools           | Requirement brainstorming, SRS/documentation assistance, code scaffolding, debugging and test-generation support. |
| React.js / Next.js       | Component-based web application development.                                                                      |
| Kotlin / Android         | Native Android application development.                                                                           |
| Node.js / Express.js     | Backend API and business-service development.                                                                     |
| Python / FastAPI         | AI/ML services and analytical model integration.                                                                  |
| Scikit-learn / XGBoost   | Structured machine-learning models for forecasting/prediction where appropriate.                                  |
| LangGraph / LangChain    | Agent orchestration and tool-based Agentic AI workflows.                                                          |
| PostgreSQL               | Relational storage for business, transaction and audit data.                                                      |
| Firebase Cloud Messaging | Potential push-notification service.                                                                              |
| REST / JSON / HTTPS      | Standard secure communication between clients, backend and integrations.                                          |

The final technology selection may be refined during design and implementation based on feasibility, course requirements, API availability, cost and testing results.

## 14. Glossary

| Term               | Definition                                                                                  |
|--------------------|---------------------------------------------------------------------------------------------|
| SME                | Small and Medium-sized Enterprise.                                                          |
| Agentic AI         | AI capable of multi-step task execution through reasoning, tool use and controlled actions. |
| Agent Orchestrator | Component that routes tasks among specialized agents and tools.                             |
| Human-in-the-loop  | Human review/approval of important AI-generated decisions.                                  |
| B2B                | Business-to-Business.                                                                       |

|             |                                                               |
|-------------|---------------------------------------------------------------|
| KPI         | Key Performance Indicator.                                    |
| CRUD        | Create, Read, Update and Delete.                              |
| LLM         | Large Language Model.                                         |
| RBAC        | Role-Based Access Control.                                    |
| Audit Trail | Recorded history of important actions.                        |
| Forecast    | Estimate of future values based on available historical data. |

## 15. Appendix A — Open Design Decisions

| No. | Decision / Risk       | Planned Handling                                                             |
|-----|-----------------------|------------------------------------------------------------------------------|
| 1   | Exact LLM provider    | Select after comparing cost, availability, latency, privacy and feasibility. |
| 2   | Forecasting model     | Start with a validated baseline and improve if data supports it.             |
| 3   | Mobile scope          | Prioritize essential operations; expand after API/core stability.            |
| 4   | B2B marketplace scale | Prototype supplier discovery first; full marketplace is future scope.        |
| 5   | Production deployment | Academic deployment first; production hardening is future work.              |
| 6   | AI autonomy           | Keep high-impact purchasing/financial actions behind approval.               |

## 16. Appendix B — SRS Structure Note

The structure and requirement style of this document were informed by the supplied Nagorik Sheba SRS v0.1, which uses a formal cover, contents, overall description, detailed functional requirements with IDs and priorities, interface requirements, non-functional requirements, architecture and appendices. [filecite@turn2file0L32-L69](#)

Its functional requirements use explicit IDs and a Priority column; this SRS adopts the same convention while expanding each BizMind BD functional module into sub-requirements. [filecite@turn2file0L192-L251](#)

Its architecture section documents both a layered system view and a step-by-step processing flow; BizMind BD follows the same professional documentation principle for its client, application, intelligence/agent, data and governance layers. [filecite@turn2file0L360-L408](#)

Document status: Version 0.1 is the initial requirements baseline. Requirements may be refined after teacher feedback, Week 2 design work and prototype validation. Major changes should be recorded through subsequent versions.